Silicon Sampling on CES 2025: Where We Stand

Two branches, one benchmark · prompting vs fine-tuning · what holds up

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In brief. Both branches try to reproduce how 15 Canadian subgroups (5 regions × 3 age groups) answered 10 CES 2025 questions, using only data from before 2025. Fine-tuning works much better than prompting (18 vs 30 answers out of 100 in the wrong place). But **a four-year-old poll, CES 2021 reused unchanged, still beats both (15 out of 100)**. Both branches find the same reason: models get the direction of subgroup differences right but shrink them to about a third of their real size.

1. The two branches, in plain terms

The common test. We take 10 questions from the 2025 Canadian Election Study (CES 2025) and 15 groups of Canadians (5 regions × 3 age groups). For each group and question, we compare how the model's fake respondents answered with how real Canadians answered (N = 20,180). Nothing from CES 2025 is shown to the models: it serves only as the answer key.

The score: total variation (TV). The share of answers that would have to move to another option to match reality. TV = 0.18 means 18 answers out of 100 are in the wrong place. Lower is better.

Both branches follow the same three steps. They differ in what happens at each one.

	Justin: prompting	Hubert: fine-tuning
Step 1. Teach the model before any test	**No teaching.** An off-the-shelf model (Qwen 3.5, 9 billion parameters), used as is.	**The model studies real survey answers.** Llama 3.3 (70 billion parameters) is shown 50,000 real answers from 8 Canadian surveys (CES 2019 and 2021, Democracy Checkup 2019 to 2024). Each example is one real person: their profile and 7 of their answers, then their answer to one more question, which the model learns to predict.
Step 2. Create fake respondents who answers the test	**1,500 real people from CES 2021** (100 per group, picked at random). Each fake respondent is a copy of one of them, as they were in 2021.	**15,000 real people from Democracy Checkup 2024** (1,000 per group, picked to match the population). Each fake respondent is a copy of one of them, as they were in 2024.
Step 3. Ask the questions what the model reads	A short description of the person: region, age, gender, province, education, income, language, religion, born in Canada, **party they identify with, left–right position, 2019 vote**. Then all 10 questions at once.	A short description of the person: age, gender, education, province, income, language. Then **their own answers to 7 questions**: left–right, party, rich–poor gap, bilingualism, trust, ties with the US, equal rights. Then one CES 2025 question at a time.
Scale	1,500 requests (10 answers each)	150,000 requests (1 answer each)
Temperature	1.0 for both: the model's natural randomness, neither reduced nor amplified.	

What one request looks like (fine-tuning branch, shortened).

```
You are a respondent to an opinion survey conducted in 2025.
Age: 18 to 24 · Gender: Man · Province: British Columbia · Income: $100,000 and over …
Your answers to other questions in the survey:
- Left-right, 0 to 10 → 9
- In federal politics, you usually think of yourself as a → Conservative
- … (5 more)
Question: How much should the federal government spend on affordable housing?
Options: Spend less / Spend about the same as now / Spend more / Don't know
```

Justin's request is similar, but the person's details are a bulleted list (including party and 2019 vote) and the 10 questions come together.

Two differences to keep in mind. The fine-tuned model has more to go on: a person's real answers to related questions, and a 2024 snapshot instead of 2021. Its better score therefore mixes two things, the training and the richer description. And with 10 times more fake respondents per group, its scores carry less random noise.

Dropped along the way (tested, not worth keeping):

- *Bigger model* (Qwen 2.4T): worse than the 9B in every condition; it piles onto neutral answers.
- *Shorter descriptions* (demographics only, without party or left–right): party information is the largest single gain for prompting.
- *Changing the 7 questions per question* (picking those closest in meaning to the one asked): worse than always using the same 7, at every training size.
- *More training examples* (100,000 instead of 50,000): no better.
- *Stretching the model's group differences* (Justin's IRT correction, ×3): gains 0.01 overall and hurts 4 of 10 questions. Even fed the real 2025 national figures, it does worse (0.098) than CES 2021 gaps (0.064).

2. Scoreboard

Mean TV over the 150 subgroups, and over the 10 national distributions. Green is closer to reality, red further. Grey rows use CES 2025 itself and only show what is reachable.

Source	Branch	By subgroup	National
Human floor (two halves of CES 2025)	Reference	0.062	0.062
Real 2025 national for every subgroup	Reference	0.085	0
CES 2021, same subgroup, unchanged	Past poll	0.151	0.141
CES 2021 regression (same descriptions as Justin)	Past poll + statistics	0.175	0.150
Fine-tuned Llama 3.3 70B (50k examples)	Hubert	0.178	0.157
Prompted Qwen 3.5 9B (full description)	Justin	0.302	0.258
Llama 3.3 70B before teaching (step 1)	Hubert	0.325	0.296

- **Fine-tuning roughly halves the error** of the same model before teaching (0.325 → 0.178) and does much better than the best prompting (0.302).
- **A fine-tuned model is about as good as a standard regression fitted on 2021** (0.178 vs 0.175 by subgroup), but neither beats simply reusing CES 2021.

3. By question

Mean TV over the 15 subgroups. **Bold** marks the best of the three forecasts on each question.

Question	CES 2021 unchanged	Fine-tuned Llama 70B	Prompted Qwen 9B	Floor
Spending: affordable housing	**0.08**	0.10	0.13	0.05
Satisfaction with democracy	**0.06**	0.12	0.23	0.05
Medical assistance in dying	**0.09**	0.14	0.23	0.09
Jobs before environment	**0.13**	0.15	0.31	0.10
Spending: defence	0.31	**0.17**	0.24	0.04
Pipelines	**0.16**	0.17	0.31	0.10
Spending: national childcare	**0.09**	0.18	0.31	0.06
Personal finances, past year	**0.13**	0.21	0.21	0.04
Spending: reconciliation	**0.17**	0.24	0.44	0.05
Immigration levels	0.31	**0.30**	0.60	0.04
Mean	**0.151**	0.178	0.302	0.062

- **CES 2021 wins on 8 of 10 questions.** The fine-tuned model wins on **defence**, where opinion moved sharply after 2021 and its training covers surveys up to 2024. On **immigration** both are poor.
- **The fine-tuned model beats prompting on every question**, by the widest margin on immigration (0.30 vs 0.60) and reconciliation (0.24 vs 0.44).
- **Shared misses.** Immigration: 62% of Canadians want fewer immigrants; prompted models chose it 2–4% of the time, and Azure's content filter removed most immigration items from fine-tuning. Reconciliation: prompted models say "spend more" 68–91% of the time, against 22% of Canadians.

4. What both branches find

- **Subgroups are flattened.** Hubert measures model subgroup gaps at about 31% of their real size (correlation 0.42; CES 2021: 69%, 0.71). Justin finds the same with an item response model: right direction on every question, slope at roughly a third of the human one.
- **Most of the error is the national level.** For fine-tuning and CES 2021 alike, the national error (0.157, 0.141) is close to the subgroup error. Giving CES 2021 gaps the real 2025 national figure drops the error to 0.064, almost the human floor (0.062).
- **On repeated questions, a past poll is the bar, and it is high.** All 10 questions were asked in CES 2021. Silicon sampling should add value on questions with no past version.

5. Next steps

1. **Report one score.** Mean TV by subgroup, for every run and baseline.
2. **Test the realistic hybrid:** national level from the fine-tuned model + subgroup gaps from CES 2021. This is the forecast behind the 0.064 ceiling, and it needs no new runs.
3. **Build a novel-question benchmark:** CES 2025 items never asked in 2019 or 2021, scored separately.
4. **Fine-tune without Azure's filter** (another host) to recover immigration.
5. **Test temperature** below and above 1.0, to see whether it causes part of the flattening.
6. **Repeat the best recipe with a new seed** before treating 50k vs 100k as settled.

Comparability: Justin's subgroup scores use 100 synthetic respondents per subgroup (Hubert: 1,000), which adds a few points of sampling noise; his scores are weighted by population (Hubert: equal weights; 0.174 vs 0.178 for the fine-tuned model). Logit and prompting figures as reported in Justin's notes of October 12–14.